

Photosynthesis Basics

Photosynthesis is the process plants use to convert sunlight into chemical energy.

Chlorophyll in plant leaves absorbs light energy and drives the synthesis of glucose.

Plants take in carbon dioxide from the air and water from the soil during photosynthesis.

Oxygen is released as a byproduct and supports respiration for many living organisms.

The light-dependent reactions occur in the thylakoid membranes inside chloroplasts.

The Calvin cycle fixes carbon dioxide into organic molecules in the stroma of chloroplasts.

Photosynthesis supports nearly all life on Earth because it produces food and oxygen.